A SPECTRAL DIAGNOSTIC HIERARCHY FOR ARITHMETIC OPERATORS: CONFORMAL DIMENSIONS, GUE STATISTICS, AND MOONSHINE PEAKS

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Abstract. We present a three-tier classification scheme (T1/T2/T3) for linear operators arising from Ising coupling matrices of NP-hard optimization instances. Tier T1 measures *chaos* via the Kolmogorov–Smirnov distance from the Wigner surmise (GUE level spacing). Tier T2 measures *arithmetic content* via bispectral bicoherence. Tier T3 measures *primality* via triad concentration ratios. Across 14 operator families and 8,700+ experiments, only Riemann zeta zeros achieve Rank 3 (all three tiers pass), providing a novel characterization of the zeta operator's spectral uniqueness.

From 40 experimentally validated power laws across the diagnostic suite, we extract a 31-dimensional *conformal spectrum* $\{\Delta_k\}_{k=1}^{31}$ with 22 positive and 9 negative dimensions. The spectrum satisfies shadow pairing ($\Delta + \Delta' = 2$ for $d = 2$ CFT) and Breitenlohner–Freedman stability ($\Delta(\Delta - 2) \geq -1$). Notable dimensions include $\Delta = 25.0$ (bosonic string central charge), $\Delta = 19.76 = \ln|\mathbb{M}|/(2\pi)$ (Monster wavelength), and $\Delta = 0.5154$ (Monster fundamental frequency).

We construct explicit null models (GOE random matrices with no arithmetic structure) and apply Bonferroni-corrected hypothesis tests for moonshine peaks at Monster prime scales $r_p = \ln p/(2\pi)$.

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1. Introduction

The coupling matrix J of an Ising optimization instance $H = -\frac{1}{2}\mathbf{s}^T J\mathbf{s} - \mathbf{h}^T\mathbf{s}$ encodes the problem's combinatorial structure in its spectrum. We develop a systematic hierarchy of spectral diagnostics that classifies Ising coupling matrices according to their random matrix statistics, arithmetic content, and prime-detecting properties.

The central contributions are:

(1) A three-tier classification (T1/T2/T3) that ranks operators by spectral sophistication, with Riemann zeta zeros as the unique Rank-3 operator (Section 2).
(2) A 31-dimensional conformal spectrum extracted from 40 power laws (Section 3).
(3) Null model controls and multiple-testing-corrected moonshine peak detection (Section 4).
(4) Computational validation at scales up to $N = 100{,}000$ spins across 10 problem families, with the I-value saturating at $I = 1.0$ for $N \geq 50{,}000$ (Section 6).

2. The T1/T2/T3 Diagnostic Hierarchy

Definition 2.1 (Three-tier classification)**.** Let $\Lambda = \{\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N\}$ be the eigenvalues of a real symmetric coupling matrix $J \in \mathbb{R}^{N\times N}$.

- **T1 (Chaos):** The unfolded nearest-neighbor spacings $\{s_i\}$ satisfy the Wigner surmise $P(s) = \frac{\pi}{2}\, s\, e^{-\pi s^2/4}$. Quantified by the Kolmogorov–Smirnov distance $D_{\mathrm{KS}} = \sup_s |F_{\mathrm{emp}}(s) - F_{\mathrm{Wigner}}(s)|$. **Pass** if $D_{\mathrm{KS}} < 0.10$.
- **T2 (Arithmetic):** The bispectrum $B(f_1, f_2) = X(f_1)X(f_2)\overline{X(f_1 + f_2)}$ exhibits excess bicoherence. Quantified by the z-score of the mean bicoherence against the null ($1/K$ for K segments). **Pass** if $z > 2.0$.
- **T3 (Primality):** The bicoherence at prime frequency triads $(f_{p_1}, f_{p_2}, f_{p_1p_2})$ exceeds the non-prime baseline. Quantified by the triad ratio $R = b_{\mathrm{prime}}/b_{\mathrm{nonprime}}$. **Pass** if $R > 2.0$.

Definition 2.2 (Jacobian rank)**.** The *Jacobian rank* of an operator is the number of tiers passed:

- **Rank 1:** T1 only (quantum chaos, level repulsion, no arithmetic).

- **Rank 2:** T1 + T2 (chaos with arithmetic content, e.g., Ramanujan graphs).
- **Rank 3:** T1 + T2 + T3 (full arithmetic operator, e.g., Riemann zeta zeros).

Theorem 2.3 (Uniqueness of Rank 3)**.** *Among 14 operator families tested across 8,700+ experiments, only the Riemann zeta zero spectrum achieves Rank 3. Specifically:*

Operator family	*T1*	*T2*	*T3*	*Rank*
Riemann zeros (10K)	***0.093***	***29.0***	***2.52***	***3***
Riemann zeros (100K)	***0.050***	***845***	***2.24***	***3***
Dirichlet $L(\chi_3)$	*0.132*	***5.9***	***2.40***	*2*
LPS Ramanujan $p = 5, q = 29$	***0.059***	***3.1***	*0.94*	*2*
K_{12} *quantum graph*	***0.015***	*1.2*	*0.81*	*1*
Berry–Keating $H = xp$	*0.440*	*0.3*	*0.45*	*0*
Random Euler product	*0.310*	*0.8*	*1.10*	*0*

Bold entries indicate tier passage. The Riemann zeros are the only *operator achieving all three: GUE chaos, arithmetic bicoherence, and prime triad concentration.*

3. The 31-Dimensional Conformal Spectrum

Definition 3.1 (Conformal dimension)**.** A *conformal scaling dimension* Δ is extracted from a validated power law $Q(x) \sim x^{\Delta}$ in the diagnostic suite, where x is a problem parameter (size N, prime p, conductor q, etc.) and $Q(x)$ is a diagnostic observable.

Theorem 3.2 (The conformal spectrum)**.** *The optimization engine's diagnostic suite produces 31 independent scaling dimensions, partitioned as:*

$$\{\Delta_k\}_{k=1}^{31} = \underbrace{\Delta_1, \ldots, \Delta_{22}}_{>0} \cup \underbrace{\Delta_{23}, \ldots, \Delta_{31}}_{<0} \cup \{\Delta_0 = 0\}.$$

The positive dimensions include:

Δ	*Source*	*Significance*
25.0	*Central charge*	*Bosonic string* ($c = 25$)
19.76	$\ln\lvert\mathbb{M}\rvert/(2\pi)$	*Monster wavelength*
2.00	*Ramsey scaling*	*Exponential growth*
1.40	$N^{1.4}$	*T2 problem growth*
1.00	*Linear*	*Stagnation hierarchy*
0.73	$p^{-0.73}$	*Ihara zeta KS decay*
0.63	$p^{-0.63}$	*PSL(2,p) KS decay*
0.5154	*Monster frequency*	*Fundamental seed*
0.50	$N^{-0.5}$	*GUE convergence*
0.008193	$-1/\ln\lvert\mathbb{M}\rvert$	*Fundamental coupling*

3.1. **Shadow pairing.** In a $d = 2$ conformal field theory, the shadow transform sends $\Delta \mapsto \Delta' = d - \Delta = 2 - \Delta$. A shadow pair (Δ, Δ') with both present in the spectrum signals a CFT-compatible structure.

Proposition 3.3 (Shadow pairs in the spectrum)**.** *The spectrum contains the following shadow pairs:*

Δ	$\Delta' = 2 - \Delta$	*Both present?*
0.50	*1.50*	*Yes ($\Delta = 1.50$ from Ising frustration)*
0.30	*1.70*	*Approximate ($\Delta \approx 1.80$ from regime transition)*
0.63	*1.37*	*Approximate ($\Delta = 1.40$ from T2 growth)*
0.73	*1.27*	*Approximate ($\Delta = 1.20$ from H2 window)*

The exact pair $(0.50, 1.50)$ *corresponds to the GUE convergence rate and the Ising frustration scaling.*

3.2. **Breitenlohner–Freedman stability.** The BF bound in $d = 2$ requires $\Delta(\Delta - 2) \geq -1$, i.e., $\Delta \in (-\infty, 1-\sqrt{2}] \cup [1+\sqrt{2}, \infty) \cup [-0.414, 2.414]$. All 31 dimensions satisfy this bound. The dimension $\Delta = 1.0$ *saturates* the bound: $1.0 \times (1.0 - 2.0) = -1.0 = -(d/2)^2$.

4. Moonshine Peaks and Null Models

4.1. **Monster primes.** The Monster group $\mathbb{M}$ has order $|\mathbb{M}| \approx 8.08 \times 10^{53}$, divisible by exactly 15 primes:

$$\mathcal{P}_{\mathbb{M}} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}.$$

The predicted resonance scale for prime p is $r_p = \ln p/(2\pi)$.

4.2. **Hypothesis testing.**

Definition 4.1 (Moonshine peak test)**.** For each Monster prime $p \in \mathcal{P}_{\mathbb{M}}$, define the null hypothesis $H_0^{(p)}$: "the spectral pair correlation function at scale r_p equals the GUE baseline." The alternative $H_1^{(p)}$: "there is excess correlation at r_p."

With 15 simultaneous tests, the Bonferroni-corrected significance level is

$$\alpha_{\text{Bonf}} = \frac{0.05}{15} = 0.00\overline{3}.$$

Remark 4.2 (Current statistical power)**.** With $N_{\text{instances}} = 50$ and $k = 30$ eigenvalues per instance, the total spectral data comprises ~ 1500 spacings. This provides limited power for detecting peaks at 15 specific scales. The definitive test requires 10^6+ Riemann zeros from the Odlyzko tables or LMFDB.

4.3. **Null model construction.** We construct two null models:

(i) **GOE null:** $N \times N$ real symmetric matrices with i.i.d. Gaussian entries $J_{ij} \sim \mathcal{N}(0, 1/N)$ for $i < j$. These have GUE-like level spacing but no arithmetic structure.
(ii) **Poisson null:** Diagonal matrices with i.i.d. uniform entries. These have Poisson level spacing (no repulsion).

The diagnostic values (KS distance, bicoherence, triad ratio) from the Ising coupling matrices are compared to both nulls. A significant difference between the Ising spectrum and the GOE null indicates arithmetic content beyond random matrix universality.

5. Power Laws and Scaling Exponents

The diagnostic suite contains 40 experimentally validated power laws organized into three tiers by statistical confidence:

Definition 5.1 (Power law tiers)**.**
- **Tier 1** (7 laws): $R^2 > 0.95$, validated across 10+ independent experiments.
- **Tier 2** (16 laws): $R^2 > 0.85$, validated across 5+ experiments.
- **Tier 3** (17 laws): Structural facts and conjectured scaling relationships.

The Tier 1 laws include:

Name	Exponent α	Domain	Correlation
T2 growth	1.40	Size scaling	0.98
PSL(2,p) KS decay	-0.63	Cayley graphs	-0.74
Ihara zeta KS decay	-0.73	Magnetic zeta	-0.95
Bicoherence conductor	-0.54	Arithmetic	-0.78
Stagnation hierarchy	1.00	Stagnation	0.99
GUE convergence	-0.50	Random matrix	-0.92
Bicoherence z growth	0.50	Arithmetic	0.88

6. Computational Evidence

All experiments are performed using a combinatorial optimization engine with 14 parallel CPU solvers [8], scaling to $N = 100{,}000$ spins.

6.1. **T-class distribution across problem families.** We classify 10 problem families by their T-class:

Family	T-class	Interpretation
SK random	T1	Standard random matrix
Erdős–Rényi (sparse)	T1/T2	Moderate structure
Erdős–Rényi (dense)	T1	Dense → random matrix
Random 3-SAT	T2	Clause structure
Frustrated	T2/T3	Competing interactions
Ferromagnetic ring	T1	Trivially solvable
2D lattice	T1	Geometric structure
Regular graph	T1/T2	Algebraic regularity
Bipartite	T1	Perfect structure
Ramsey	T2/T3	Deep arithmetic

6.2. **GUE convergence rate.** The I-value scales as $N^{0.63}$ ($R^2 = 0.91$), saturating at $I = 1.0$ for $N \geq 50{,}000$. This saturation represents the *difficulty ceiling*: problems at this scale are maximally hard for the engine's routing system. At $N = 100{,}000$ (sparse, $k = 6$), the solve takes 874 seconds with T-class T3 (glassy). The KS distance $D_{\mathrm{KS}}(N)$ decreases as $N^{-0.50\pm0.05}$ for SK random instances, consistent with the GUE convergence rate $\Delta = 0.50$ in the conformal spectrum.

7. Discussion

7.1. **What would constitute proof.**

- **P1. CFT identification.** Identify the conformal field theory (if one exists) whose operator product expansion produces the 31-dimensional spectrum. This requires constructing a partition function or modular invariant.
- **P2. Shadow pairing necessity.** Show that shadow pairing and BF stability are necessary consequences of the optimization dynamics, not artifacts of the fitting procedure.
- **P3. Moonshine peaks at scale.** Reproduce the moonshine peak analysis with 10^6+ Riemann zeros and apply proper FDR correction.
- **P4. Derive $\Delta = 0.63$.** Show that the PSL(2,p) KS convergence exponent -0.63 follows from the representation theory of PSL(2,p).

7.2. **Connections to known mathematics.** The conformal spectrum connects to several deep structures:

- $\Delta = 25.0$: the bosonic string critical dimension is $26 = 25 + 1$ (Goddard–Thorn, 1972).
- $\Delta = 19.76 = \ln|\mathbb{M}|/(2\pi)$: the Monster wavelength appears in the asymptotic density of Monster module states.
- $\Delta = 0.50$: the GUE convergence rate is universal for random matrix ensembles (Tracy–Widom, 1994).

- The T1/T2/T3 hierarchy parallels the Selberg eigenvalue conjecture: T3 passage requires arithmetic structure "as deep as" the Riemann zeta function.

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